FRE-GY 9743 — Assignment 3

Bond Futures: Conversion Factor, Implied Repo and the Cheapest-to-Deliver

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Repository (computational part: cashflow products and display visitor):
https://github.com/Yuri48/FRE-GY-9743-Assignments-3/tree/Yuri_Moghaddam_Branch
Every number below is reproduced by `hw3_bond_future_numbers.py` in the same repository.

Setup and notation

Prices are per 100 face. S is the clean spot price, $\mathrm{AI}(t)$ the accrued interest, $S^d = S + \mathrm{AI}(0)$ the dirty spot price, T_d the delivery date and τ the repo accrual factor to T_d. Repo is a money-market simple rate; three months is $\tau = 0.25$ (90 days on ACT/360). Buying the bond today with repo financing and delivering it at T_d gives the forward price

$$\mathrm{BF}^d = S^d(1 + r\tau) - \sum_{t_i \le T_d} c_i\big(1 + r(T_d - t_i)\big), \qquad \mathrm{BF} = \mathrm{BF}^d - \mathrm{AI}(T_d),$$

where BF is the *clean* forward and the sum runs over coupons paid before delivery.

The bond in Question 1 has exactly 9 years to maturity, a 4% semi-annual coupon and $S = 100$. Today is therefore a coupon date: $\mathrm{AI}(0) = 0$, $S^d = 100$, the yield is 4%, and the next coupon is six months away, after delivery.

1. Normalized forward price

1) Three-month forward at $r_{\mathbf{mkt}} = 2\%$. No coupon is paid before T_d, so

$$\begin{aligned} \mathrm{BF}^d &= 100\,(1 + 0.02 \times 0.25) = 100.5000, \\ \mathrm{AI}(T_d) &= 100 \times \tfrac{0.04}{2} \times \tfrac{3}{6} = 1.0000, \\ \mathrm{BF} &= 100.5000 - 1.0000 = \mathbf{99.5000}. \end{aligned}$$

Equivalently, the clean forward is spot minus carry. Over three months the bond earns 1.00 of coupon (4%×0.25) and costs 0.50 to finance (2%×0.25). Carry is +0.50, so BF = 100−0.50 = 99.50. The forward is below spot because the coupon yield (4%) exceeds the repo rate (2%).

2) Conversion factor. With 18 semi-annual periods, $v = 1/1.03$ and $c/y = 0.04/0.06$:

$$K = \frac{1}{100}\Big[\,2\sum_{i=1}^{18} v^i + 100\,v^{18}\Big] = \frac{c}{y} + \Big(1 - \frac{c}{y}\Big)v^{18} = 0.666667 + 0.333333 \times 0.587395 = \mathbf{0.862465}.$$

The bond pays 4%, below the 6% notional coupon, so it trades below par at 6% and $K < 1$.[1]

[1] The CME computes K from the bond's remaining maturity on the first day of the delivery month, rounded down to a whole quarter. That would be 8.75 years here, giving $K = 0.865306$ and $\mathrm{BF}/K = 114.988$. I use the assignment's

3) Normalized forward.

$$\frac{\mathrm{BF}}{K} = \frac{99.5000}{0.862465} = \mathbf{115.3670}.$$

4) What the conversion factor does.

- **It puts every deliverable bond on a common scale.** Bonds in the basket have different coupons and maturities, so their raw prices cannot be compared: at the same yield, a 2% bond might trade at 80 and a 6% bond at 100. K_x is the price per unit face that bond x would have if it yielded 6%. It says how many units of the hypothetical 6% notional bond one unit of bond x is worth. Dividing by K_x expresses each bond's forward in units of the notional bond, so a single futures price F covers the whole basket. Delivering one unit face of bond x is paid $F\,K_x$ plus accrued interest.
- **The normalization is exact only at 6%.** If the curve were flat at 6%, $S_x/K_x = 100$ for every bond (this holds exactly in the three-bond example of Question 3), and the short would be indifferent between deliverable bonds apart from carry. At any other yield the scaling is imperfect, which is why a cheapest-to-deliver bond exists (Question 3).
- **Accrued interest (the hint).** K is a ratio of *clean* prices, so it rescales only the clean part of the price. Accrued interest is a known cash amount owed to the seller for the coupon earned since the last payment. It is not rescaled; it is paid in full on top:

  $$\text{invoice}_x = F\,K_x + \mathrm{AI}_x(T_d).$$

  The forward therefore has to be converted to clean before it is divided by K, which is why $\mathrm{AI}(T_d) = 1.00$ was subtracted in part 1). Dividing the dirty forward instead gives 100.50/0.862465 = 116.53, which effectively pays $1.00/K = 1.16$ of accrued instead of 1.00. That error would also vary across the basket with each bond's position in its coupon cycle: just after a coupon AI ≈ 0, just before one AI $\approx c/2$. Working with clean prices removes coupon timing from the comparison, and the accrued interest is passed through unchanged.

2. Cheapest-to-deliver

1) Implied repo at $F = 115.3670$. Set $\mathrm{BF}(S, r^{\mathrm{imp}}, T_d)/K = F$. With no coupon before delivery this is $FK + \mathrm{AI}(T_d) = S^d(1 + r^{\mathrm{imp}}\tau)$, so

$$\begin{aligned} r^{\mathrm{imp}} &= \frac{1}{\tau}\Big[\frac{FK + \mathrm{AI}(T_d)}{S^d} - 1\Big] = \frac{1}{0.25}\Big[\frac{115.3670 \times 0.862465 + 1.0000}{100} - 1\Big] \\ &= \frac{1}{0.25}\Big[\frac{100.5000}{100} - 1\Big] = \mathbf{2.00}\% \end{aligned}$$

(1.99994% before rounding F to four decimals). The implied repo equals the market repo because the quoted futures price is exactly this bond's normalized forward at 2%. A cash-and-carry with this bond earns exactly the funding rate, which is what we expect if it is the CTD and there is no arbitrage. If a coupon c_i fell at $t_i < T_d$, the formula would become $r^{\mathrm{imp}} = \big[FK + \mathrm{AI}(T_d) + \sum c_i - S^d\big]/\big[S^d\tau - \sum c_i(T_d - t_i)\big]$.

Implied repo is very sensitive to the futures price over short horizons: $\partial r^{\mathrm{imp}}/\partial F = K/(S^d\tau) = 3.45\%$ per point, so one 32nd of a point moves it by about 10.8 bp.

definition (9 years); it is the one consistent with the futures price quoted in Question 2.

2) Choosing the CTD from implied repos. r_x^{imp} is the return on a cash-and-carry with bond x: buy it spot (pay S_x^d), sell the future, hold to T_d and deliver it for $FK_x + \text{AI}_x(T_d)$. Because BF_x increases in r and $F = \min_x \text{BF}_x(r_{\text{mkt}})/K_x$,

$$F \le \frac{\text{BF}_x(r_{\text{mkt}})}{K_x} \;\; \forall x \quad \Longrightarrow \quad r_x^{\text{imp}} \le r_{\text{mkt}} \;\; \forall x,$$

with equality for the bond that attains the minimum. Hence

$$\textbf{CTD} = \arg\max_{x \in \mathcal{X}} r_x^{\text{imp}},$$

the bond with the highest implied repo, which should be close to r_{mkt}. Following the hint, the short (the party that delivers) reacts to $r_{\text{mkt}} \neq r^{\text{imp}}$ as follows:

- $r_x^{\text{imp}} > r_{\text{mkt}}$: borrow at r_{mkt} in repo, buy bond x, sell the future and deliver. This locks in $r_x^{\text{imp}} - r_{\text{mkt}}$ without risk. Buying the bond and selling the future pushes S_x up and F down until $r_x^{\text{imp}} = r_{\text{mkt}}$, so in equilibrium the highest implied repo is about r_{mkt}.
- $r_x^{\text{imp}} < r_{\text{mkt}}$: delivering x earns less than the cost of financing it. The short would rather deliver another bond, or simply lend cash at r_{mkt}. The further r_x^{imp} is below r_{mkt}, the more expensive x is to deliver.

So I would rank the basket by implied repo, take the highest as today's CTD, and compare it with the term repo actually available for that bond: the GC rate, or the bond's own special rate if it trades special. An equivalent price measure is the net basis $\text{BF}_x(r_{\text{mkt}}) - FK_x$; the CTD has the smallest. In practice the highest implied repo sits slightly *below* GC because the short owns the delivery options (quality/switch, timing, end-of-month, wild card). The futures price therefore trades a little under $\min_x \text{BF}_x/K_x$, and the CTD's net basis is positive by roughly the value of those options. Example from Question 3(b) at a flat 6.25% yield: implied repos are A 1.59%, B 2.00%, C 1.51%, so B is the CTD.

3. Scenario analysis

Can the CTD at $t+N$ be predicted? Yes, conditional on a scenario. The conversion factors are fixed for the life of the contract, so the CTD at $t+N$ is

$$\text{CTD}(t+N) = \arg\min_x \; \frac{\text{BF}_x\big(S_x(t+N),\, r(t+N),\, T_d\big)}{K_x}.$$

For each scenario of the curve at $t+N$ (level, slope, repo, specialness), reprice every deliverable bond, recompute the normalized forwards or implied repos with the remaining carry, and take the minimum. A grid of scenarios, or a simulation of curve factors, gives a distribution of the CTD. From it we get the probability of a switch and the value of the short's delivery option. The main drivers:

1. **Yield level relative to the 6% notional coupon.** To first order

$$S_x(y)/K_x = 100\, P_x(y)/P_x(6\%) \approx 100\, e^{D_x(6\%-y)},$$

where D_x is duration. Below 6% this ratio rises with duration, so the *lowest-duration* bond (short maturity, high coupon) is cheapest. Above 6% it falls with duration, so the *highest-duration* bond (long maturity, low coupon) is cheapest. At exactly 6% all ratios equal 100 and carry decides.

2. **Curve slope.** A steepening (long yields up relative to short) cheapens the long bonds and moves the CTD toward longer maturities. A flattening moves it toward shorter maturities.
3. **Distance from a switch point.** When normalized forwards are far apart, the CTD is stable. When they are close (yields near 6%, or bonds of similar duration), small moves change it.
4. **Carry, repo and supply.** Higher-coupon bonds have more positive carry when coupons exceed repo, which lowers their forwards. Carry shrinks as T_d approaches, so $\mathrm{BF}_x \to S_x$ and the CTD at delivery is $\arg\min_x S_x(T_d)/K_x$. A bond trading special in repo, a squeeze on the CTD, a new issue entering the basket or a bond aging out of it can all change the answer.

Example. Hypothetical basket, repo 2%, 3 months to delivery, flat curve unless twisted. Bonds are A (5.75%, 7y), B (4%, 9y; the bond from Question 1) and C (2%, 10y). Their conversion factors are $K = 0.9859,\ 0.8625,\ 0.7025$ and their modified durations at 4% are 5.8, 7.5 and 8.8.

BF/K at flat yield	3%	4%	5%	6%	7%	8%
A 5.75% 7y	**118.063**	**111.280**	**104.950**	**99.042**	93.524	88.367
B 4.00% 9y	124.498	115.367	107.004	99.341	92.313	85.864
C 2.00% 10y	130.077	118.965	108.904	99.788	**91.524**	**84.028**

The CTD (bold) is the short, high-coupon bond A up to a 6.19% flat yield, then B up to 6.34%, then the long, low-coupon bond C. At exactly 6%, A is still cheapest because of its higher carry. At a 6.25% level, a twist around the 9-year point is enough to move the CTD:

Slope (bp per year of maturity)	-10	-5	0	$+5$	$+10$
CTD at a 6.25% level	A	A	B	C	C
CTD at a 4% level	A	A	A	A	A

At 4% the CTD stays A even under a $+25$ bp/year twist (75 bp between the 7y and 10y points). Near 6%, five basis points per year are enough to switch it.

Why we care which bond is delivered in the end.

- **Pricing.** The futures price follows the CTD: $F \approx \mathrm{BF}_{\mathrm{CTD}}/K_{\mathrm{CTD}}$ minus the value of the delivery options. Getting the CTD wrong means mispricing F and the basis.
- **Hedging.** The futures DV01 is roughly the CTD's forward DV01 divided by K_{CTD}. To hedge bond y we use $h = \mathrm{DV01}_y\, K_{\mathrm{CTD}}/\mathrm{DV01}_{\mathrm{CTD}}$ contracts. If the CTD switches from A (duration 5.8) to C (duration 8.8), the futures' duration jumps by about 50%, and a static hedge becomes badly sized. Because F behaves like the minimum of several bond prices, the future is negatively convex.
- **The delivery option and basis trading.** The short holds the switch option and the long has sold it. A long-basis position (long bond, short futures) is long this option, and its P&L depends on whether, and when, the CTD switches.
- **Physical settlement.** The short must source the bond it delivers, which creates repo specialness and squeeze risk on the CTD. The long receives a specific bond (the one that is cheapest for the short, i.e. the least valuable to the long) with its own coupon and duration. The cash paid, $FK_x + \mathrm{AI}_x(T_d)$, depends on which bond arrives.